

SMART INDIA HACKATHON (SIH 2026) SYSTEM SPECIFICATION

SafeWay V3

Smart Indoor Navigation & Crowd-Aware Emergency Evacuation System

SafeWay V3 is a real-time IoT hardware and software safety platform combining indoor graph navigation, ESP32 hazard & crowd detection, multi-tier crowd balancing (corridors, zones, exits), life-safety elevator fire policies, and centralized security control.



Multi-Floor Turn-by-Turn
Dijkstra / A* navigation with  Step-Free Accessibility support.



Sequential Evacuation Flow
[USER] → [SAFE CORRIDOR] → [EXIT] → [ASSEMBLY]



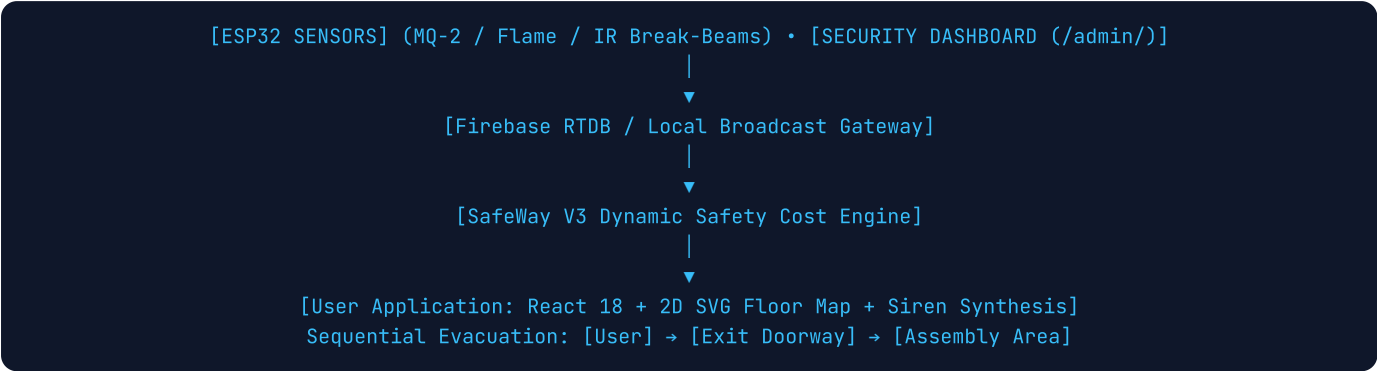
ESP32 IoT Node Firmware
MQ-2 Smoke, Optical Flame, and Dual IR Break-Beam Occupancy Counting.



Area of Rescue Assistance
2-hour fire-rated safe havens for wheelchair users during elevator fire lockouts.

1. System Architecture & Mathematical Engine

1.1 End-to-End System Workflow



1.2 Dynamic Safety Cost Formula

$Cost(e) = Distance(e) + P_blocked + P_hazard + P_crowd + P_accessibility + P_policy$

- $P_blocked = \infty$ (Locked / collapsed paths)
- $P_hazard = \infty$ (Active fire) | +50 (Smoke / heat)
- $P_crowd = +60$ (High crowd) | +20 (Medium crowd) | +0 (Low crowd)
- $P_access = \infty$ (Accessibility violation e.g. wheelchair on stairs)
- $P_policy = \infty$ (Ordinary elevator locked down during active fire)

1.3 QR Indoor Positioning Dictionary

QR Code Key	Location Name	Floor	Assigned Zone
SW-ENT-MAIN	Main Entrance	Floor 1	Zone A (Lobby)
SW-CAFE	Cafeteria & Dining	Floor 1	Zone C (Dining)
SW-LAB101	Science Lab 101	Floor 1	Zone B (Labs)
SW-R00M201	Room 201 Lecture Hall	Floor 2	Zone D (West Classrooms)
SW-R00M204	Room 204 Faculty Lounge	Floor 2	Zone D (West Classrooms)
SW-LIBRARY	Central Library	Floor 2	Zone E (East Wing)

2. Hardware Integration & Verification (18 / 18 Tests Passed)

2.1 ESP32 Directional Occupancy Counting Logic

Using dual IR break-beam sensors placed at doorway thresholds:

- **Beam 1 (Entry) broken then Beam 2 (Exit) broken:** Person Entered ($\text{Occupancy} + 1$).
- **Beam 2 (Exit) broken then Beam 1 (Entry) broken:** Person Exited ($\text{Occupancy} - 1$).
- **Crowd Thresholds:** \$0-7\$ Low • \$8-19\$ Medium • \$20+\$ High.

2.2 Automated Test Suite Results

Scenario 1: Main Entrance → Room 204 Navigation Normal mode calculates 73m shortest path correctly.	100% PASSED
Scenario 2: Multi-Floor Elevator Navigation Vertical transition uses central lift in normal mode.	100% PASSED
Scenario 3: Fire Hazard Avoidance Zone B fire completely rejects Exit 1 and detours to Exit 3 / Exit 2.	100% PASSED
Scenario 4: Multi-Tier Dynamic Crowd Balancing Diverts evacuee away from congested East corridor and Exit 1.	100% PASSED
Scenario 5: Closed Exit Door Fallback Closed Exit 2 is rejected; next best reachable exit is selected.	100% PASSED
Scenario 6: Sequential Assembly Route Path strictly terminates at open-air designated assembly grounds.	100% PASSED
Scenario 7: Step-Free Accessibility Mode Zero stair edges used in step-free wheelchair routing.	100% PASSED
Scenario 8: Fire Elevator Policy & Refuge Area Elevators disabled during fire; routes upper-floor wheelchair user to Area of Refuge.	100% PASSED

3. SIH 6-Stage Demonstration Script & Execution

Stage	Action	System Behavior
Stage 1	User scans <code>SW-R00M204</code> → Searches Main Entrance	Calculates 73m shortest path via Elevator in Normal Mode.
Stage 2	Admin activates Emergency in <code>/admin/</code>	User app transitions immediately into red evacuation mode with audio siren.
Stage 3	ESP32 in Zone B detects smoke	SafeWay marks Zone B as fire danger; invalidates Exit 1.
Stage 4	Exit 2 crowd surges to High	SafeWay dynamically applies <code>+\$60\$</code> crowd penalty to Exit 2.
Stage 5	Dynamic Route Selection	SafeWay evaluates all exits and recommends Exit 3 (North-West Wing).
Stage 6	Evacuation Execution	User safely exits building to Assembly Area B (West Courtyard).

3.1 Quick Start Commands

```
# Run automated test suite
npm test

# Launch local server
npm start
```